A General Variable Neighborhood Search for Logistic Network Flow Design with Nonlinear Edge Costs and Transshipment Capacities

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Abstract. We study the problem of routing multiple commodities through a logistic transport network so as to minimise the joint cost of vehicle movements and of transshipment (reload) operations at intermediate storages. The problem combines two features that are common in practice but rarely treated together: edge costs that are *nonlinear* in the transported volume, because freight is carried by vehicles of fixed capacity, and *hard upper bounds* on the volume that can be reloaded at each node. We formulate the problem as a mixed integer linear program in both an arc-based and a path-based form, derive its linear relaxation, and propose a metaheuristic solver based on General Variable Neighborhood Search (GVNS). The method works in the path space and relies on three problem-specific neighborhoods that generate new candidate paths by (i) shortest paths under a reload-aware metric, (ii) edges carrying partially loaded vehicles, and (iii) detours around congested storages. We evaluate the method on our open benchmark [18] for this problem: a corpus of realistic logistic networks calibrated on real road graphs of nine European countries, five U.S. states and the European part of Russia, with near-planar topology (Zipf city sizes, a spatial-network budget model, and a doubly constrained gravity demand model). On 35 synthetic and 15 real-geometry instances with up to 150 storages, GVNS returns solutions within about one percent of the best dual bound on small networks—a branch-and-bound lower bound substantially tighter than the LP relaxation—is stable across random seeds, and for a comparable wall-clock budget is up to 55% better than a time-limited exact solver on the medium instances.

Keywords: multi-commodity flow · network design · variable neighborhood search · metaheuristics · mixed integer programming · logistics

1 Introduction

Organising commodity flows between distribution centres is one of the central problems of logistic networks. The growth in scale and complexity of modern

warehousing structures creates a persistent demand for efficient algorithms that build optimal transportation plans. Two characteristics make realistic instances hard. First, the cost of using a road is *nonlinear* in the volume transported along it: goods travel in vehicles of fixed capacity, so the relevant quantity is the integer *number of vehicles*, obtained by rounding the carried volume up to the capacity. Second, every storage has a limited *transshipment capacity*—an upper bound on the total volume that may be unloaded, stored and reloaded there. Together these features turn an otherwise classical multi-commodity flow problem into a mixed integer program whose feasible region is shaped by node capacities and whose objective contains a step-wise vehicle term.

Exact methods for multi-commodity flow problems are computationally expensive because the problem is NP-hard. Column generation [1] is a classical approach that solves path-based formulations by dynamically building a set of promising paths, and is frequently combined with branch-and-price. In practice, heuristics and metaheuristics—greedy algorithms, genetic algorithms, tabu search, simulated annealing and local-search-based methods—are widely used to obtain good solutions in acceptable time [2]. Among them, Variable Neighborhood Search (VNS) and its variants have proven effective across a broad range of optimisation problems. Masri et al. [3] applied a multi-start VNS to the single-path multi-commodity flow problem, and General VNS has performed well on the capacitated vehicle routing problem and other flow-constrained logistic problems [4]. Closer to the present setting, problem-specific heuristics have proven effective on real-life logistic routing problems, such as the truck-and-trailer routing problem with time windows and split deliveries [20,21].

Contributions. The aim of this work is the development and study of a metaheuristic algorithm, based on General Variable Neighborhood Search (GVNS), for distributing commodity flows in a logistic network with nonlinear edge costs and limited storage capacities. Specifically, we

1. formalise the problem and give equivalent arc-based and path-based MILP models together with their linear relaxation (Section 2);
2. design a GVNS that searches in the space of commodity paths and is equipped with three problem-specific neighborhoods that exploit reload costs, partial vehicle loads and storage congestion (Section 3);
3. report a computational study on our open benchmark [18] (Section 4), comparing GVNS against the LP relaxation, its rounded version, and MIP dual bounds and incumbents from an exact solver (Section 5).

The main methodological elements are the specialised GVNS neighborhoods adapted to the path-based formulation and its constraints, together with the exact-solver study that quantifies the quality of the resulting solutions.

2 Problem statement and model

2.1 Verbal statement

We are given a transport network of storages connected by roads. Prescribed volumes of goods must be moved from origins to destinations so as to minimise the total cost of moving vehicles along roads plus the cost of transshipment (unloading, storing and reloading) at intermediate transit storages. Each storage has a limited transshipment capacity and a per-unit reload cost. Freight travels in vehicles of fixed capacity, and each road has a cost per vehicle pass; the number of vehicles on a road must be sufficient to carry the total volume routed through it. The problem belongs to the class of multi-commodity flow problems with additional node-capacity constraints and a vehicle-induced nonlinear edge cost, and admits two equivalent formulations: arc-based and path-based.

2.2 Data

The road network is a directed graph $G = (V, E)$, where V is the set of storages and E the set of roads. We are given a set of transportation requests (commodities) K; for each $k \in K$ we know an origin $s_k \in V$, a destination $t_k \in V$, and a demand $d_k \in \mathbb{R}_+$ to be shipped from s_k to t_k. For each storage $v \in V$ we know the per-unit reload cost $f_v \in \mathbb{R}_+$ and the maximum admissible transshipment volume $W_v \in \mathbb{R}_+$. Each edge $e \in E$ has a per-vehicle traversal cost $c_e \in \mathbb{R}_+$. Finally, $C \in \mathbb{R}_+$ is the capacity of one vehicle.

2.3 Decision variables

Arc-based formulation. Let $x_e^k \in \mathbb{R}_+$ be the flow of commodity $k \in K$ on edge $e \in E$ and $y_e \in \mathbb{Z}_+$ the number of vehicles traversing e. The model has $|K| \cdot |E| + |E|$ variables.

Path-based formulation. Let Π_k be the set of all paths from s_k to t_k. Let $x_p^k \in \mathbb{R}_+$ be the flow of commodity k on path $p \in \Pi_k$, and $y_e \in \mathbb{Z}_+$ as above. The model has $\sum_{k \in K} |\Pi_k| + |E|$ variables.

2.4 Objective

Denote by $\delta^-(v) \subseteq E$ and $\delta^+(v) \subseteq E$ the sets of edges entering and leaving v. The objective minimises the sum of vehicle movement cost (first term) and transit reload cost (second term).

Arc-based:

$$\min \left(\sum_{e \in E} c_e \, y_e \; + \sum_{k \in K} \sum_{\substack{v \in V \\ v \neq s_k,\, v \neq t_k}} \sum_{e \in \delta^-(v)} f_v \, x_e^k \right). \tag{1}$$

Path-based:

$$\min \left(\sum_{e \in E} c_e \, y_e \; + \sum_{k \in K} \sum_{p \in \Pi_k} \sum_{\substack{v \in p \\ v \neq s_k,\, v \neq t_k}} f_v \, x_p^k \right). \tag{2}$$

2.5 Constraints

(1) Demand satisfaction.

$$\text{arc:} \quad \sum_{e\in\delta^+(v)} x_e^k - \sum_{e\in\delta^-(v)} x_e^k = \begin{cases} d_k, & v = s_k \\ -d_k, & v = t_k \\ 0, & \text{otherwise} \end{cases} \qquad \forall k \in K,\ \forall v \in V; \tag{3}$$

$$\text{path:} \quad \sum_{p\in\Pi_k} x_p^k = d_k, \qquad \forall k \in K. \tag{4}$$

(2) Vehicle capacity on edges. Enough vehicles must run on each edge to carry the total flow:

$$\text{arc:} \quad \sum_{k\in K} x_e^k \le C\, y_e, \qquad \forall e \in E; \tag{5}$$

$$\text{path:} \quad \sum_{k\in K} \sum_{p\in\Pi_k:\, e\in p} x_p^k \le C\, y_e, \qquad \forall e \in E. \tag{6}$$

(3) Storage transshipment capacity.

$$\text{arc:} \quad \sum_{\substack{k\in K \\ v\neq s_k,\, v\neq t_k}} \sum_{e\in\delta^-(v)} x_e^k \le W_v, \qquad \forall v \in V; \tag{7}$$

$$\text{path:} \quad \sum_{\substack{k\in K \\ v\neq s_k,\, v\neq t_k}} \sum_{p\in\Pi_k} \sum_{v\in p} x_p^k \le W_v, \qquad \forall v \in V. \tag{8}$$

2.6 Classification and linear relaxation

The objective and all constraints are linear, and the variables are mixed: the vehicle counts y_e are integer while the flows x_e^k (or x_p^k) are continuous. The problem is therefore a *mixed integer linear program* (MILP), an NP-hard class for which no polynomial-time exact algorithm is known in general.

A standard way to bound the optimum is the LP relaxation: dropping the integrality of y_e turns the MILP into a linear program solvable in polynomial time, and its optimum is a lower bound (for minimisation). Once y_e is continuous, the capacity constraints (5)–(6) hold with equality at optimum, so y_e can be eliminated:

$$y_e = \frac{1}{C} \sum_{k\in K} x_e^k \quad \text{(arc)}, \qquad y_e = \frac{1}{C} \sum_{k\in K} \sum_{p\in\Pi_k:\, e\in p} x_p^k \quad \text{(path)}. \tag{9}$$

Substituting (9) into the objective yields a relaxed problem in the flow variables only. In the path-based form,

$$\min \sum_{k\in K} \sum_{p\in\Pi_k} x_p^k \left(\frac{1}{C} \sum_{e\in p} c_e + \sum_{\substack{v\in p \\ v\neq s_k,\, v\neq t_k}} f_v \right), \tag{10}$$

which makes explicit that an optimal relaxed solution prefers paths with low travel cost and few transit nodes.

3 The GVNS solution approach

3.1 General Variable Neighborhood Search

VNS [5,6] is a metaheuristic whose core idea is the systematic change of neighborhood structures while exploring the solution space. Given a problem $\min\{f(x) : x \in X\}$ and a family of neighborhoods $N_k(x),\ k = 1, \ldots, k_{\max}$, the classical scheme alternates three steps: a *shake* (a random move in N_k to escape local optima), a *local search*, and a *neighborhood change* (on improvement, accept the move and reset $k \leftarrow 1$; otherwise advance to $k+1$). When the local search is itself a Variable Neighborhood Descent (VND)—a deterministic sweep through the neighborhoods ordered from intensification to diversification—the method is called General VNS (GVNS), which often outperforms single-neighborhood heuristics [7,8]. Our instantiation searches the space of commodity paths; its concrete form is Algorithm 2, built on the neighborhood-processing primitive of Algorithm 1.

3.2 Preprocessing and initial solution

Transport networks are typically sparse directed graphs in which not every pair of storages is joined by a direct road, and not every transit node needs a reload. Working on the raw graph both shrinks the feasible region (through node capacities) and may even render the instance infeasible if no path exists between an origin and a destination.

To avoid this, the graph is *completed*: for every pair of storages v, u we add (or update) an edge whose cost equals that of the shortest path from v to u in G. Each edge of the resulting complete graph thus represents the cheapest stop-free route between its endpoints. This guarantees the existence of a feasible solution, because flow routed on the direct origin–destination edge does not pass through any transit node and hence violates no node-capacity constraint. From the relaxed objective (10) it follows that the cheapest, transit-free direct edges are exactly the ones a good solution prefers. We therefore take as the initial solution the relaxed LP solution that sends each commodity entirely along its direct origin–destination edge, with vehicle counts rounded up:

$$y_e = \left\lceil \frac{1}{C} \sum_{k \in K} x_e^k \right\rceil, \qquad \forall e \in E. \tag{11}$$

3.3 Neighborhoods

Since the vehicle counts y_e are determined directly from the flows via (11), the neighborhoods vary the flows. Predicting feasible per-edge flows is hard because edges are shared by many commodities and adjacent flows are strongly coupled, so the whole network must be considered at once. A single path, by contrast, describes the routing of just one commodity, and adding a new path to an already feasible solution never worsens it. The neighborhoods therefore *build and modify*

the set of candidate paths for each commodity: for every commodity a set of new paths is generated and added to the model, a MILP solver is run for a limited time to redistribute flow, and—if the objective improves—all new paths carrying nonzero flow are kept. Three neighborhoods define how the new paths are generated.

N1 – Shortest paths (reload-aware). Classical shortest-path search ignores reload costs. We instead build an auxiliary complete graph G' whose edge weights measure the per-unit cost of moving one unit of commodity, including reloads at the endpoints:

$$c'_{vu} = f_v + f_u + \frac{c_{vu}}{C}, \qquad \forall (v,u) \in E. \tag{12}$$

Before GVNS starts, a pool of shortest paths under c' is precomputed for each commodity; new paths are then drawn from this pool, either at random or preferring low-cost paths. The neighborhood parameters are the number of new paths per commodity and a boolean choosing the selection strategy. During VND all combinations are swept (one to three new paths, both strategies); in Shake, five random paths are always added, which helps escape local optima.

N2 – Partially loaded edges. A partially loaded vehicle has spare room: routing extra flow over such an edge can free another vehicle and reduce the vehicle count. For every edge we compute the residual load

$$load_e = \Big(\sum_{k \in K} \sum_{p \in \Pi_k \colon e \in p} x_p^k \Big) \bmod C, \qquad \forall e \in E, \tag{13}$$

and build a graph G'' that favours edges with free capacity. Fully loaded edges ($load_e = 0$) are least attractive, since new flow would require an extra vehicle, and keep their weight from G'. The remaining edges receive

$$c''_{vu} = f_v + f_u + \frac{load_{vu}}{C} \Big(\frac{c_{vu}}{C} \Big)^{\alpha}, \qquad \alpha \le 1, \tag{14}$$

so that smaller residual load (more free room) lowers the weight and raises priority; α tunes how strongly travel cost influences the weight. Shortest-path search on G'' then tends to build paths through partially loaded vehicles. The number of new paths is set dynamically: a counter of fully loaded paths and an accumulator of total free room stop generation once a threshold is reached (in VND, at most two fully loaded paths or total free room exceeding one vehicle capacity; in Shake, three paths and $3C$).

N3 – Detours around congested storages. Storages whose transshipment volume is at or near their capacity are bottlenecks, and routing around them is desirable. For the current solution we compute a congestion ratio

$$overload_v = \frac{1}{W_v} \sum_{\substack{k \in K \\ v \neq s_k,\, v \neq t_k}} \sum_{p \in \Pi_k} \sum_{v \in p} x_p^k, \qquad \forall v \in V. \tag{15}$$

Algorithm 1: `ProcessNeighborhood`: enlarge the path pool, re-optimise the flows by MILP, and keep the move iff it improves the objective.

Function `ProcessNeighborhood`(P, x, z, N, *div*, τ)**:**
  `//` incumbent path pool P, flows x, objective $z = f(x)$; updated in place
  $Q \leftarrow \bigcup_{k \in K}$ `GenPaths`(N, k, *div*) `//` candidate paths; N selects the rule
  if $Q = \emptyset$ **then return false**
  $(x', z') \leftarrow$ `SolveMILP`($P \cup Q$, x, τ) `//` redistribute flow; warm start x, limit τ
  if $z' < z$ **then**
    $P \leftarrow P \cup \{ p \in Q : x'_p > 0 \}$; $x \leftarrow x'$; $z \leftarrow z'$
    prune from P every path carrying zero flow `//` keep the model bounded
    return true
  else
    discard Q
    return false

Storages are sorted by decreasing $overload_v$, and a subset of the most congested ones is selected according to a threshold (and trimmed or padded to a target size). For each prefix $\{v_1, \ldots, v_i\}$ of the ordered set, those nodes (except the commodity's own origin and destination) are deleted from G' and a shortest path is recomputed, yielding detours that bypass the busiest nodes and shift flow to less loaded ones. In VND the subset size ranges from $|V|/5$ to $|V|/4$, with threshold 0.75 and one path per configuration; in Shake the range widens to $|V|/4$–$|V|/3$, the threshold drops to 0.5 and three paths are generated per configuration.

3.4 Overall algorithm

The method is given by Algorithms 1 and 2. Algorithm 1 is the primitive shared by the three neighborhoods: it generates a set of candidate paths—the only step that differs between N_1, N_2 and N_3, via `GenPaths`—enlarges the path pool, re-optimises the commodity flows with the path-based MILP of Section 2, warm-started from the current solution, and, *only* if the objective improves, commits the new paths that carry flow while pruning those that do not, which keeps the model bounded. Algorithm 2 embeds this primitive in the GVNS template: starting from the rounded-LP solution of Eq. (11), each round shakes with the current neighborhood—diversified parameters and a larger MILP budget τ_{shk}—and then runs the VND sweep $N_1 \to N_2 \to N_3$ with a smaller budget τ_{vnd}, since VND explores a narrower region. The neighborhood index resets on improvement

Algorithm 2: `GVNS` for logistic network flow design.

Input : graph $G = (V, E)$, commodities K, vehicle capacity C, time limit $t_{\max}$
Output: path pool P with commodity flows x

build the reload-aware complete graph G'; precompute a shortest-path pool per commodity
$P \leftarrow \{\,(s_k, t_k) : k \in K\,\}$; $(x, z) \leftarrow$ `RoundedLP(`P`)` `// initial solution, Eq.` (11)
$k \leftarrow 1$
while $k \leq 3$ **and** CpuTime $< t_{\max}$ **do**
  `ProcessNeighborhood(`$P,\ x,\ z,\ N_k,\ \mathbf{true},\ \tau_{\text{shk}}$`)` `// shake: diversified, larger budget`
  $z_0 \leftarrow z$
  repeat
    $imp \leftarrow$ `ProcessNeighborhood(`$P,\ x,\ z,\ N_1,\ \mathbf{false},\ \tau_{\text{vnd}}$`)`
    if $\neg\, imp$ **then** $imp \leftarrow$ `ProcessNeighborhood(`$P,\ x,\ z,\ N_2,\ \mathbf{false},\ \tau_{\text{vnd}}$`)`
    if $\neg\, imp$ **then** $imp \leftarrow$ `ProcessNeighborhood(`$P,\ x,\ z,\ N_3,\ \mathbf{false},\ \tau_{\text{vnd}}$`)`
  until $\neg\, imp$ **or** CpuTime $\geq t_{\max}$
  if $z < z_0$ **then** $k \leftarrow 1$ **else** $k \leftarrow k + 1$ `// neighborhood change`
return $(P,\ x)$

and advances otherwise; a global limit $t_{\max}$ bounds the run, and the search stops once a full sweep brings no improvement.

4 Benchmark instances

We evaluate GVNS on our open benchmark [18,19] for this problem, published as a dataset on Zenodo (CC-BY-4.0); we summarise here only what is needed for the study and refer the reader to [18] for the full generator. The corpus targets upper-level road networks connecting regional sorting centres, with a generator calibrated on OpenStreetMap [9] data for nine European countries, five large U.S. states and the European part of Russia. It reproduces the structural statistics of real regional transport systems—Zipf-distributed city sizes [10], a spatial-network budget topology [11], centrality-driven transshipment capacities [12,13] and a doubly constrained gravity demand model [14]—yielding *near-planar, large-diameter* graphs (mean vertex degree about 3.575), the opposite of small-world networks. The corpus comprises 35 synthetic instances (10 with 10 storages, 7 each with 20 and 30, 5 with 50, and 3 each with 100 and 150) and 15 real-geometry instances built on the calibration road graphs (11–90 storages); we use all 50, each exactly reproducible from its published seed and parameters.

Table 1: Main GVNS and exact-solver settings. VND/Shake values are given as VND / Shake where they differ.

Setting	Value
Neighborhoods $k_{\max}$	3 (N_1, N_2, N_3)
GVNS time limit	$2\lvert V\rvert$ minutes
Random seeds	3
Vehicle capacity C	50
N_1 new paths per commodity (VND / Shake)	1–3 / 5
N_2 travel-cost exponent α	0.5
N_2 free-room stop (VND / Shake)	$1C$ / $3C$
N_3 congestion threshold (VND / Shake)	0.75 / 0.5
N_3 storage-subset size (VND / Shake)	$\lvert V\rvert/5$–$\lvert V\rvert/4$ / $\lvert V\rvert/4$–$\lvert V\rvert/3$
Exact-solver MIP-gap target	0
Exact-solver budget	30 min ($\lvert V\rvert$=10) … 4 h ($\lvert V\rvert$=150)

5 Computational experiments

5.1 Setup, baselines and metric

All experiments were run on the HPC cluster "cHARISMa" [17] (nodes with two AMD EPYC CPUs, 44 cores and ≥720 GB RAM per node, Rocky Linux 9). Each job was pinned to eight cores. The exact models were solved with HiGHS 1.14.0 [15] and, as an independent cross-check of the lower bounds, with SCIP 10.0.2 [16]; the metaheuristic uses HiGHS as its in-loop redistribution solver, as before. GVNS is stochastic (random path choices in the shake step), so every instance was solved with three independent random seeds and we report the best objective together with its across-seed mean and standard deviation. Unless stated otherwise the exact solves were run to a target MIP gap of zero (i.e. to optimality where possible) under a size-dependent wall-clock budget (30 min for $\lvert V\rvert = 10$, growing to 4 h for $\lvert V\rvert = 150$); GVNS is run with a time limit of twice the number of storages, in minutes. The two budgets are comparable—at $\lvert V\rvert = 50$, for example, GVNS is allowed 100 min against the exact solver's 2 h, and at $\lvert V\rvert = 10$ it is 20 min against 30 min.

The main algorithm settings are collected in Table 1. The size of the arc-based MIP grows steeply with $\lvert V\rvert$—from about 6 000 variables at $\lvert V\rvert = 10$ (edges $\lvert E\rvert = 90$, commodities $\lvert K\rvert = 67$) to 3.7×10^8 at $\lvert V\rvert = 150$ ($\lvert E\rvert = 22\,350$, $\lvert K\rvert = 16\,762$)—which is why the exact model becomes intractable well before that size while GVNS, working in the much smaller path space, remains applicable throughout. A GVNS run took, on average, from about 640 s at $\lvert V\rvert = 10$ to 3.7×10^3 s at $\lvert V\rvert = 50$ and 1.8×10^4 s at $\lvert V\rvert = 150$.

Lacking known optima, we compare GVNS against four references:

- **LP relaxation** – the optimum of the relaxed problem (Section 2.6); a *lower bound*. On the completed graph it is attained by routing every commodity along its direct edge (Section 3), so it has a closed form and is available at every size, though—ignoring vehicle integrality—it is loose;

- **Rounded LP** – the relaxed solution with y_e rounded up; this is the GVNS starting solution and a valid *upper bound*;
- **MIP dual bound** – the best lower bound proved by the exact branch-and-bound solver within its budget, i.e. the global lower bound of the search tree: the LP relaxation strengthened by branching and cutting planes, which rises towards the optimum as the tree is explored. It is thus a *valid* lower bound ($\underline{z} \leq z^*$) that—unlike the root LP relaxation, which ignores vehicle integrality—is typically tight (within a few percent of the optimum); it is the key reference, letting us bound the true optimum even when optimality is not proved;
- **MIP incumbent** – the best feasible solution the exact solver returns within a budget comparable to the GVNS run time.

Because the true optimum is generally not available, the primary quality measure we report is the *optimality gap of GVNS to the best valid lower bound*,

$$gap_{\mathrm{LB}} = \frac{Result_{\mathrm{GVNS}} - \underline{z}}{\underline{z}}, \qquad \underline{z} = \text{best MIP dual bound},$$

which upper-bounds the true optimality gap of GVNS. We additionally report the ratio of each reference objective to that of GVNS ($Ratio_{\mathrm{method}} = Result_{\mathrm{method}}/Result_{\mathrm{GVNS}}$), so values below 1 are better than GVNS and above 1 are worse.

On the hardness of the exact model. A first, and perhaps surprising, observation is that the arc-based MILP is hard even at the smallest sizes: with the vehicle-count integrality it is a large mixed-integer program (e.g. already $\sim$6000 variables at $|V| = 10$), and *no* instance – not even $|V| = 10$ – was solved to *proven* optimality within the budgets above by either solver. The branch-and-bound *dual bound*, however, is tight (within a few percent of the incumbent) and, crucially, well above the LP relaxation; it is therefore the right yardstick for judging GVNS. We also note that the two solvers behave very differently on this model: their dual bounds agree closely (mutually cross-validating the lower bound), but HiGHS finds markedly better incumbents, whereas SCIP fails to find any feasible solution on several instances with $|V| \geq 45$ and runs out of memory at $|V| = 150$. HiGHS is thus both the stronger exact solver here and the natural in-loop choice.

5.2 Results

Tables 2 and 3 summarise the synthetic and real-geometry results; Figure 2 plots the optimality gap and the reference ratios against instance size. Four consistent trends emerge.

First, GVNS is near-optimal on small instances. Its gap to the MIP dual bound is only 0.97% at $|V| = 10$ and stays around 1–2% up to $|V| \approx 30$, then grows with size (to $\sim 8\%$ at $|V| = 50$), reflecting the rapidly expanding path space. Since the dual bound is a valid lower bound, this gap bounds the true distance of GVNS from the optimum rather than comparing it against itself.

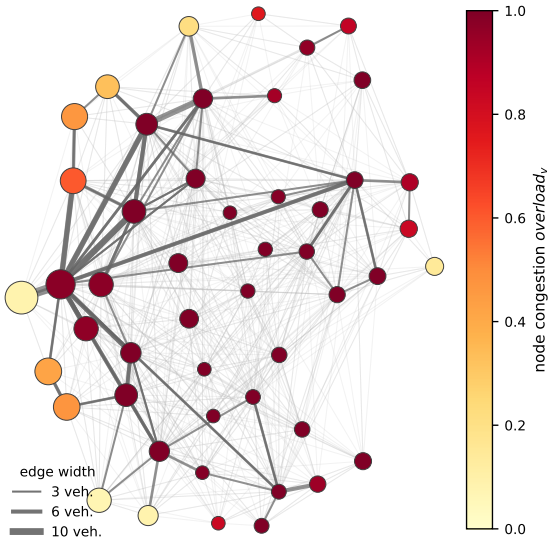
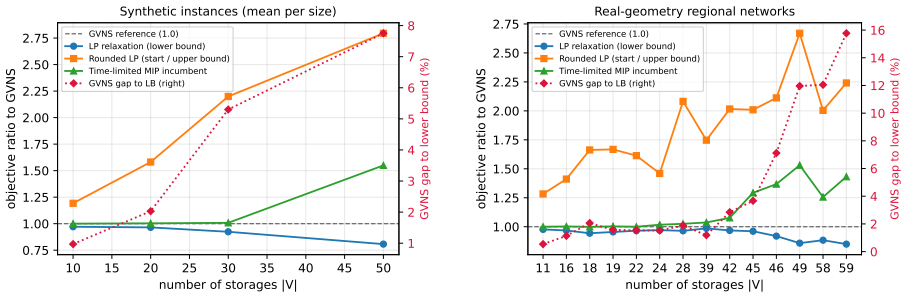


Fig. 1: A GVNS solution on the Germany real-geometry network ($|V| = 49$), drawn on its geographic layout. Node area encodes the transshipment capacity W_v and node colour the congestion $overload_v$ (dark = at capacity). All used edges are drawn faintly; backbone edges carrying at least three vehicles are emphasized with width proportional to the vehicle count.

Second, GVNS is very stable across random seeds. The across-seed standard deviation of the objective is below 0.5% of the mean on every instance size (e.g. 0.07% at $|V| = 10$, 0.31% at $|V| = 50$), so a single run is representative and the method is robust to its randomisation.

Third, GVNS matches or beats the exact solver at equal time, increasingly so with size. On small instances the time-limited MIP incumbent essentially ties GVNS ($Ratio_{\text{MIP}} \approx 1.00$ for $|V| \leq 30$); from $|V| = 45$ the exact solver falls behind sharply, returning solutions 1.3–1.55× more expensive than GVNS at $|V| = 49$–59. In other words, for a comparable wall-clock budget the exact solver is up to ∼55% worse than GVNS. We are able to obtain tight dual bounds up to $|V| = 59$; for $|V| \gtrsim 90$ the arc model reaches millions of variables and the exact branch-and-bound—both its tight dual bound and its incumbent—becomes intractable, leaving only the closed-form but loose LP-relaxation lower bound. Even there GVNS keeps improving on its rounded-LP start by large factors (2.2–3.0×; Tables 2 and 3).

Fourth, the real-geometry networks mirror the synthetic ones (Table 3): the gap to the dual bound rises from 0.5% (Denmark, $|V| = 11$) to 12–16% (Germany $|V| = 49$, Spain $|V| = 59$), and the exact incumbent degrades from parity to 1.3–1.5× GVNS over the same range. This corroborates both the fidelity of the generator and the robustness of GVNS.



(a) Synthetic instances (mean per size). **(b)** Real-geometry regional networks.

Fig. 2: Left axis: objective ratio of each reference method to GVNS versus network size (values below the GVNS line at 1.0 are better than GVNS, above are worse). Right axis (dotted): the GVNS optimality gap to the best valid lower bound. Only sizes with an exact dual bound are shown.

5.3 Neighborhood ablation

To assess the contribution of each neighborhood we ran a leave-one-out ablation: GVNS is re-run with a single neighborhood disabled (in both shake and VND) and compared, over three seeds, against the full method. Table 4 reports the mean objective increase on a representative subset of synthetic ($|V| = 20$–50) and real-geometry ($|V| = 16$–49) instances.

Three effects stand out. N_2 (vehicle consolidation over partially loaded edges) is the dominant operator: removing it costs up to 17.5% on the larger synthetic instances and 7.2% on Germany, confirming that filling spare vehicle capacity is the main lever on the objective. N_1 contributes a smaller but consistent improvement. N_3 (detours around congested storages) is *situational*: it is essentially inactive when storages are far from their limit (the synthetic and small real instances), but becomes valuable exactly when the transshipment capacity binds—on Germany, where almost every storage operates at capacity (Figure 1), disabling N_3 worsens the objective by 3.6%. Each neighborhood thus earns its place, addressing a different source of cost.

6 Conclusion

We addressed the design of optimal logistic transport flows with nonlinear, vehicle-induced edge costs and hard storage transshipment capacities. We gave arc-based and path-based MILP models and their linear relaxation, and proposed a GVNS that operates in the path space through three problem-specific neighborhoods: reload-aware shortest paths, partially loaded edges, and detours around congested storages.

On 35 synthetic and 15 real-geometry instances of up to 150 storages, GVNS produced solutions within about one percent of the best dual bound on small

Table 2: Synthetic results aggregated by network size. gap_{LB} is the GVNS optimality gap to the best MIP dual bound (a valid lower bound); $Ratio_{\bullet}$ are objective ratios to GVNS. "†" = exact model intractable within the budget (memory/time). n = number of instances.

$\lvert V\rvert$	n	gap_{LB}	Ratio LP-relax.	Ratio Rounded LP	Ratio MIP-inc.	GVNS std
10	10	0.97%	0.971	1.192	1.000	0.07%
20	7	2.03%	0.965	1.581	1.004	0.14%
30	7	5.30%	0.924	2.199	1.008	0.49%
50	5	7.75%	0.807	2.795	1.550	0.31%
100	3	†	0.541	2.186	†	0.17%
150	3	†	0.372	2.197	†	0.26%

† at these sizes the arc model has millions of variables, so the exact branch-and-bound—its tight dual bound and its incumbent—is intractable; the LP-relaxation lower bound and the rounded-LP upper bound remain available in closed form (Sections 2.6 and 3).

networks (a MIP lower bound far tighter than the LP relaxation) and, for a comparable budget, up to 55% cheaper than a time-limited exact solver on the medium ones; it is stable across random seeds, and its optimality gap grows with size, as expected from the exploding path space. A leave-one-out ablation confirmed that each neighborhood earns its place: consolidating flow over partially loaded edges (N_2) is the dominant operator, while congestion detours (N_3) become valuable specifically on capacity-tight instances. Future work includes integrating column-generation pricing to feed the path pool and validating the approach on operational data from real logistic systems.

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Table 3: Results on the real-geometry regional networks. Sizes are the storage counts of the generated instances.

Network	$\|V\|$	gap_{LB}	Ratio LP-relax.	Ratio Rounded LP	Ratio MIP-inc.	GVNS t (s)
Denmark	11	0.5%	0.977	1.283	1.000	523
Belgium	16	1.1%	0.968	1.411	1.003	1279
Netherlands	18	2.1%	0.944	1.663	0.999	2191
Hungary	19	1.6%	0.955	1.668	1.003	975
Ohio USA	22	1.5%	0.965	1.613	1.000	1535
Georgia USA	24	1.5%	0.971	1.460	1.017	1823
Illinois USA	28	1.9%	0.964	2.082	1.025	2605
California USA	39	1.2%	0.986	1.747	1.037	2409
United Kingdom	42	2.8%	0.968	2.016	1.077	3877
Sweden	45	3.7%	0.961	2.009	1.293	2818
Italy	46	7.1%	0.919	2.112	1.368	2669
Germany	49	12.0%	0.858	2.670	1.531	3292
Texas USA	58	12.0%	0.884	2.004	1.256	4168
Spain	59	15.8%	0.850	2.241	1.432	4145
Europe Russia	90	†	0.763	3.017	†	4784

† exact branch-and-bound (tight dual bound and incumbent) intractable at this size; the LP-relaxation and rounded-LP bounds remain available in closed form.

Table 4: Neighborhood ablation: mean objective increase (best of three seeds) when a single neighborhood is disabled, relative to the full GVNS.

Disabled neighborhood	Synthetic	Real
N_1 (reload-aware shortest paths)	+1.5%	+0.7%
N_2 (partially loaded edges)	+7.2%	+2.5%
N_3 (congestion detours)	+0.0%	+0.8%

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